Power Law in Deep Neural Networks: Sparse Network Generation and Continual Learning With Preferential Attachment

Fan Feng, Lu Hou, Qi She, Rosa H. M. Chan, *Senior Member, IEEE*, and James T. Kwok, *Fellow, IEEE*

***Abstract*— Training deep neural networks (DNNs) typically requires massive computational power. Existing DNNs exhibit low time and storage efficiency due to the high degree of redundancy. In contrast to most existing DNNs, biological and social networks with vast numbers of connections are highly efficient and exhibit scale-free properties indicative of the power law distribution, which can be originated by preferential attachment in growing networks. In this work, we ask whether the topology of the best performing DNNs shows the power law similar to biological and social networks and how to use the power law topology to construct well-performing and compact DNNs. We first find that the connectivities of sparse DNNs can be modeled by truncated power law distribution, which is one of the variations of the power law. The comparison of different DNNs reveals that the best performing networks correlated highly with the power law distribution. We further model the preferential attachment in DNNs evolution and find that continual learning in networks with growth in tasks correlates with the process of preferential attachment. These identified power law dynamics in DNNs can lead to the construction of highly accurate and compact DNNs based on preferential attachment. Inspired by the discovered findings, two novel applications have been proposed, including evolving optimal DNNs in sparse network generation and continual learning tasks with efficient network growth using power law dynamics. Experimental results indicate that the proposed applications can speed up training, save storage, and learn with fewer samples than other well-established baselines. Our demonstration of preferential attachment and power law in well-performing DNNs offers insight into designing and constructing more efficient deep learning.**

***Index Terms*— Complex networks, continual learning, network pruning, power law distribution.**

I. Introduction

DEEP neural networks (DNNs) have achieved state-of-the-art performance in various tasks, such as speech recognition, visual object recognition, and image classification [1]. However, network connectivity and subsequent degree distributions can be fixed by design, resulting in many redundant connections. In real-world practice, redundancy makes training DNNs time- and storage-consuming.

To address the redundancy issues in DNNs, network pruning methodologies have been proposed [2], [3], [4], [5], [6]. In particular, if insignificant connections are pruned and the remaining connections are then retrained, the resultant sparse network often suffers little to no performance degradation [3], [4], [5], [7], [8]. However, these pruning techniques are developed in a data-driven fashion, which also requires a considerable computational burden. Instead, this work aims to investigate whether we can generate sparse and well-performing DNNs with little redundancy in a computational-efficient way. To this end, we propose to identify the underlying structure pattern of the well-performing DNNs and then utilize the identified pattern to evolve optimal networks for sparse network generation and continual learning. In this way, the generation process is topology-driven, which would largely save the computational requirement compared with data-driven approaches.

The topology of DNNs can be modeled from the complex network perspective. In general, a network has both static and dynamic properties [9]. Static properties describe the network topology, while dynamic properties describe the dynamics governing the network evolution and explain how the topology is formed. In this work, we aim to identify an optimal topology inside DNNs and dynamically generate well-performing DNNs with the discovered topology using minimal computational load.

Power law distribution has been widely used to describe the underlying structures of a wide variety of physical, biological, and artificial networks [10], [11], [12]. It is well-known that the power law can originate in an evolving network via preferential attachment [10], in which new connections are preferentially made to the more highly connected nodes. Networks exhibiting a power law distribution are also called scale-free networks [13].

In this article, we investigate whether the topology of sparse DNNs also exhibits power law properties as observed in their biological counterparts, and how we can use the power law dynamics to evolve time- and storage-efficient DNNs, which can learn sequential tasks continually. As shown in Fig. 1, we first observe that, similar to the distribution of sparse connections in social and biological neural networks, the distribution of sparse connections in DNNs can be fit by the

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Fan Feng and Rosa H. M. Chan are with the Department of Electrical Engineering and the State Key Laboratory of Terahertz and Millimeter Waves, City University of Hong Kong, Hong Kong, China (e-mail: rosachan@cityu.edu.hk).

Lu Hou and James T. Kwok are with the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology, Hong Kong, China.

Qi She is with the Bytedance AI Lab, Beijing 10000, China.

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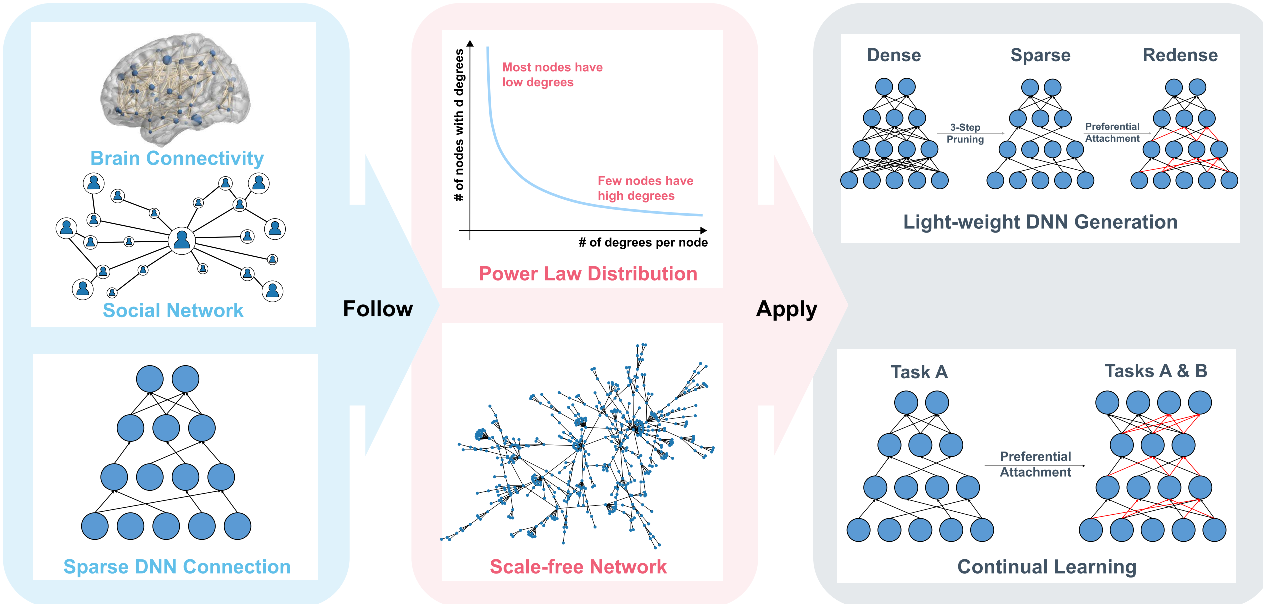

Fig. 1. Brief summary of the findings presented in this article. The distribution of the sparse connections in sparse DNNs: MLPs and CNNs (e.g., LeNet-5-like CNN, AlexNet, VGG-16, and Resnet-50) can be empirically fit by TPL distribution. Power law can be originated from preferential attachment [10]. To study the dynamic properties in DNNs, we find that, with the arrival of a new task, the new connections follow the preferential attachment process under continual-learning settings. This phenomenon is consistent with the sparse connectivity pattern discovered from neuroscience. We then utilize this property to: 1) generate lightweight DNNs with a postpruning step using preferential attachment and 2) continual learning with sequential tasks using the preferential attachment. The results have shown that the preferential attachment in both network generation and continual learning can: 1) speedup network training; 2) save storage; and 3) learn with fewer data samples.

power law distribution (see Section III). Then, we propose the preferential attachment model in DNNs and find the existence of both power law and the behavior of preferential attachment in continual learning (see Section IV). We then utilize the discovered power law topology to evolve sparse DNNs (see Section V). Specifically, we apply preferential attachment to construct lightweight networks and utilized power law dynamics in continual learning settings to learn tasks sequentially. Results show that our proposed application can: 1) speed up network training; 2) save storage; and 3) learn better with fewer data samples. In essence, our contributions are fourfold.

1) We first verify the scale-free properties in the connectivity of a variety of sparsified DNNs, ranging from multilayer perceptron (MLP) to deep convolutional neural network (CNN) models (AlexNet, VGG, ResNet, DenseNet, EfficientNet, and NF-Net).
2) To model the power law dynamics in growing DNNs, we propose the preferential attachment for DNNs, which can be identified in the continual learning process.
3) Based on the identified topology, we leverage preferential attachment to grow DNNs in sparse network generation and continual learning tasks.
4) The evaluation results indicate that the proposed novel techniques can generate well-performing DNNs with smaller network sizes, faster training speeds, and fewer samples required.

II. Related Works

A. Graph Topology in DNNs

Different from the works [14], [15], [16], which aim to design optimal network structures through a computational and empirical fashion, several studies [17], [18], [19] have investigated the links between network performance and underlying graph topologies inside the connectivity of DNNs and generated optimal DNNs in a topology-driven way. You et al. [17] explore the statistical relationship between the network performance and layerwise underlying graph topologies. Xie et al. [18] propose the RandWire method to generate DNNs using various random graph topologies. Prabhu et al. [19] propose a deep expandable network to generate DNN through layerwise bipartite graphs based on expandable graph topology. Similarly, our work aims to identify the existence of power law distribution inside the underlying topology of layerwise connectivity, which is shown as the degree distribution in DNNs. Furthermore, based on the identified topology, we propose novel applications to evolve optimal DNNs in network generation and continual learning tasks.

B. Network Pruning

Network pruning leads to efficient deep learning via reducing the model size and inference time for deep architectures [20]. The network pruning techniques can be divided into two categories, including nonstructured and structured pruning, based on the pruning granularity.

Nonstructured pruning methods aim to remove the uninformative weights. The earliest work in [2] proposes to use the second derivative of the loss function to make a balance between network complexity and training error. Recently, Han et al. [4], [5] propose the magnitude-based metric to prune unimportant weights with small absolute values. To avoid the network damaged by incorrect pruning in [4] and [5], Guo et al. [21] further incorporate a splicing step into the pruning framework to recover the incorrectly pruned weights. Instead of employing a magnitude-based approach, sparse VD [22] approximates a posterior probability distribution for each weight to dropout. The method is in line with sparse Bayesian learning [23] and recent advances in using (sparse) Bayesian learning techniques for large-scale models [24], [25], [26], [27], [28].

Instead, the structured pruning methods remove the entire filters in DNNs according to different importance metrics. Liu et al. [6] use the scaling factors in batch normalization layers to evaluate the channel importance. Li et al. [29] utilize the magnitude evaluation to compute the importance scores for channels. He et al. [30] use a LASSO regression and least square reconstruction to prune the channels. Molchanov et al. [31] utilize criteria-based pruning with fine-tuning by backpropagation for efficiently structured pruning. Yeom et al. [32], inspired by neural network interpretability, propose the layerwise relevance propagation for neuron/filter pruning.

Besides these, some meta-analysis on network pruning has been proposed recently. Frankle and Carbin [3] propose the lottery ticket hypothesis, where dense, randomly initialized, and feed-forward networks containing subnetworks can have comparable performances compared with full networks using similar training iterations. Several follow-ups extend it into various applications in deep learning (e.g., computer vision [33], natural language processing [34], [35], recommendation systems [36], and out-of-distribution learning [37]) and prove the hypothesis theoretically [38]. Le and Hua [39] identify that the learning rate schedules in fine-tuning or retraining phase play important roles in the final network performance.

In this article, we empirically observe that there exhibits power law distribution in the sparsified networks pruned by the most representative nonstructured pruning methods [3], [5]. Furthermore, utilize this finding to evolve well-performing sparse networks.

C. Continual Learning

Continual learning is a topic of interest in a wide range of real-life AI applications. In continual learning, the learning agent requires learning a sequence of tasks by memorizing the previously learned tasks and fast adaptation to future unknown tasks [40], [41], [42], [43], [44], [45], [46].

Backward and forward transfers are the learning objectives in continual learning [41], [47], [48]. The backward transfer is measured by the performance of previously seen tasks using the currently trained model. The conventional way of training a sequence of tasks will result in a low backward transfer model since the network tends to forget the previously learned knowledge representations. This phenomenon is also well-known as *catastrophic forgetting* in neural networks [49]. Forward transfer measures how the previously acquired knowledge will improve learning accuracy in future tasks.

Three research approaches have been proposed for continual learning tasks: 1) **regularization-based approaches**, including EWC [50], LwF [51], CBLN [52], FCL3 [53], and SI [54], which aim to preserve the most critical knowledge in previous tasks to alleviate catastrophic forgetting; 2) **structure-based approaches**, including PNN [55], DEN [56], and PathNet [57], which dynamically expand the network size to minimize the interference between the current tasks and previously seen tasks; and 3) **memory-based approaches** aim to preserve the previous knowledge by maintaining a replay buffer that stores either samples or feature representations from previous tasks, including real samples (e.g., ER [58], SER [59], auxiliary exemplars [60], and A-GEM [61]), generated pseudosamples from previous tasks (e.g., DGR [62], TMNs [63], RecallNet [64], and GMNet [65]), and feature replay [66]. All three major research approaches are beneficial to both forward and backward transfer in continual learning tasks. Among them, structure-based approaches explicitly build up modules for both backward and forward transfer. Specifically, there is a separate subnetwork for each task or task group. This isolation structure can minimize interference among tasks. In the meanwhile, each subnetwork has internal connections with the previous subnetworks, advancing forward knowledge transfer. Regularization- and memory-based approaches are tailored for alleviating catastrophic forgetting issues, while there are no explicit designing architectures on forward transfer.

Our work fits in the structured-based approaches. However, instead of blindly expanding the network size or dense internal connections [55], [57], we leverage the identified power law dynamics to grow the network for future tasks in a sparse and topology-driven manner. Different from other works (dynamically expandable network (DEN) [56]), which conducted sparse network growth via complicated optimization, we demonstrate a topology-driven manner, making the growth more efficient and effective.

TABLE I

SUMMARY OF VARIABLE AND FUNCTION DEFINITIONS IN SECTION II-D

Variable	Definition
x	Measurement
α	Exponent
$F(\cdot)$	Cumulative distribution function
$S(\cdot)$	Complementary cumulative distribution function
x_{min}	Lower threshold in TPL
x_{max}	Upper threshold in TPL
u_i	Random sample from uniform distribution

D. Power Law Distribution

Power law's probability density function is of the form

$$f(x) \propto x^{-\alpha} \tag{1}$$

where x is the measurement and $\alpha > 1$ is the exponent. Typically, $2 < \alpha < 3$ though there are occasional exceptions [67]. Table I gives a complete summary of all parameters and variables in power law distribution. In particular, power law can describe the phenomena where a small number of occurrences are located at the tail of a distribution.

Power law and scale-free topology have key dynamic advantages over various network topologies, including structural variety and controllable predetermined results, which are essential for the evolutionary process in networks [68]. The derived networks have already led to innovative findings [69] in the efficient social communications (social networks) [70], brain intelligence (neuroscience) [71], [72], and efficient physical embedding of networks in brains and computer circuits [73].

The cumulative distribution function (CDF) $F(x)$ and the complementary CDF (CCDF) $S(x)$ function are defined as

$$F(x) = \int_{x_{\min}}^{x} f(x)\mathrm{d}x, \quad S(x) = 1 - F(x). \tag{2}$$

The CDF and CCDF of the power law distribution are best represented in log–log scale plots. Due to the heavy tail in empirical data, it is preferable to use the log–log CCDF plot to find the exponent of a power law distribution. From (2), $\log(S(x)) = -(\alpha-1)\log(x) + (\alpha-1)\log(x_{\min})$, which shows that the log–log CCDF plot is a line with negative slope $-(\alpha-1)$.

In practice, few phenomena obey the power law exactly [67], [74]. The actual degree distribution often has a power law regime followed by a fall-off because of the finite size and temporal limitations of the collected data, and/or underlying physics constraints [75]. Such networks, sometimes called broad-scale or truncated scale-free networks [76], have been commonly observed in nature. An example is the anatomical network of the human brain [77]. Another example is that the size of forest fires is naturally limited by the availability of fuel and by climate, and the upper bounded power law fits the data better [78]. Modifications of the power law have been proposed to model this upper truncation effect [75], [79]. In this article, we focus on the truncated power law (TPL) distribution proposed in [79], which explicitly includes lower and upper thresholds.

The TPL distribution, with a lower threshold $x_{\min}$ and upper threshold $x_{\max}$, captures these effects [79]. The PDF and CDF of a TPL distribution when $\alpha \neq 1$ are [79]

$$f(x) = \frac{1-\alpha}{x_{\max}^{1-\alpha} - x_{\min}^{1-\alpha}} x^{-\alpha}, \quad F(x) = \frac{x^{1-\alpha} - x_{\min}^{1-\alpha}}{x_{\max}^{1-\alpha} - x_{\min}^{1-\alpha}}.$$

While the log–log CCDF plot for the standard power law distribution is a line, the log–log CCDF for the TPL is

$$\log\left(S(x)\right) = \log\left(x^{1-\alpha} - x_{\max}^{1-\alpha}\right) - \log\left(x_{\min}^{1-\alpha} - x_{\max}^{1-\alpha}\right). \tag{3}$$

As $\lim_{x\to x_{\max}} \log(S(x)) = -\infty$, the log–log CCDF plot for TPL has a fall-off near $x_{\max}$. Moreover, when $x_{\max}$ is large, $\log(S(x)) \simeq (1-\alpha)\log(x) - (1-\alpha)\log(x_{\min})$, and the log–log CCDF plot reduces to a line. When $x_{\max}$ gets smaller, the linear region shrinks, and the fall-off starts earlier. Let $\tilde{y} = \log\left(S(x)\right)$ and $\tilde{x} = \log(x)$; then (3) can be rewritten as

$$\tilde{y} = \log\left(10^{(1-\alpha)\tilde{x}} - x_{\max}^{1-\alpha}\right) - \log\left(x_{\min}^{1-\alpha} - x_{\max}^{1-\alpha}\right). \tag{4}$$

When x only takes integer values (instead of continuous values), the probability function of the TPL distribution becomes [80]

$$f(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min}, x_{\max})} \tag{5}$$

where $\zeta(\alpha, x_{\min}, x_{\max}) = \sum_{k=x_{\min}}^{x_{\max}} k^{-\alpha}$. The corresponding CCDF is: $S(x) = (\zeta(\alpha, x, x_{\max})/\zeta(\alpha, x_{\min}, x_{\max}))$.[1]

III. Truncated Power Law Observed in Deep Models

In this section, we aim to identify the existence of the TPL distribution inside the connectivity of sparse DNNs. Furthermore, we explore the link between DNN performances and the power law fitting goodness.

A. Truncated Power Law Fitting

Here, we introduce the TPL fitting in sparse DNNs. It was previously unknown whether the degree distributions in artificial neural networks follow the power law. The connectivity of the network is often fixed by design and not learned from data. Meanwhile, many network connections are redundant. Hence, we report the findings from networks that have been sparsified, in which only important connections are kept.

We obtain sparsified networks via the representative non-structured network pruning methods [3], [5]. Among them, a dense network is first trained. For each layer, a fraction of $s \in (0, 1)$ connections with the smallest magnitudes is pruned. The unpruned connections are then retrained. To avoid potential performance degradation, connections to the output layer are always left unpruned, as are common in network pruning [29] and continual learning research [57]. Despite there exists another pruning category, structured pruning, the granularity of the most reported structured pruning is shown as channels, groups, and layers [6], [29], [30], [81]. However, if structured pruning is conducted, only nodes (or feature maps in CNNs) and layers will be pruned, while the degrees of the remaining nodes are still unchanged. Thus, we cannot evaluate TPL in the structurally pruned DNNs.

DNNs are of finite size, and each node can only make limited connections. This suggests that upper truncation and the TPL are more appropriate for modeling connectivity. We define either each neuron in MLP or each channel in CNN as the node. Hence, the degree of each node is the number of nonzero weights connected to that neuron or channel. In particular, we will use the discrete TPL, as the degree is integer-valued. Moreover, different layers can differ greatly in the number of nodes and the maximum number of connections that each node can make. Thus, the nature of features extracted at different layers is also different. We fit a TPL distribution to every layer separately because it is more appropriate to study connectivity in terms of layers.

We have followed [67] to estimate $x_{\min}$, $x_{\max}$ and α. Specifically, $x_{\min}$ and $x_{\max}$ are chosen by minimizing the difference between the probability distribution of the observed data and the best-fit power law model as measured by the Kolmogorov–Smirnov (KS) statistic

$$D = \max_{x\in\{x_{\min}, x_{\min}+1,\ldots,x_{\max}-1, x_{\max}\}} |S(x) - P(x)|$$

where $S(x)$ and $P(x)$ are the CCDFs of the observed data and fit power law model for $x \in \{x_{\min}, x_{\min}+1, \ldots, x_{\max}-1, x_{\max}\}$.

[1]Mean and variance of TPL distributions are analyzed in Supplementary Material A.

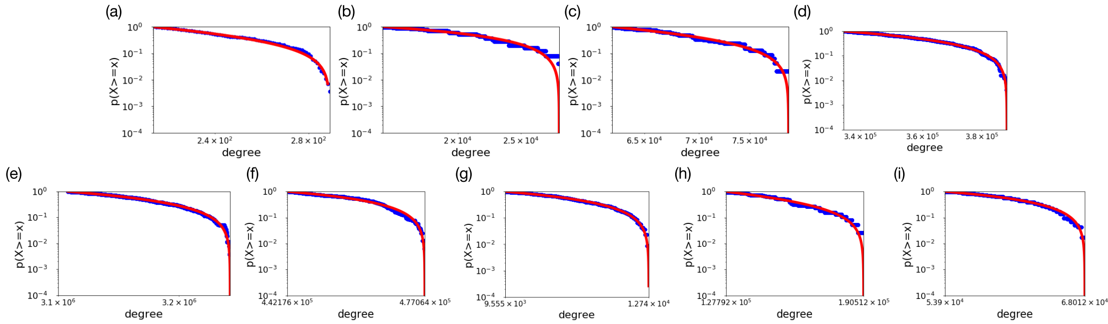

Fig. 2. Log–log CCDF plot (blue) and TPL fit (red) for layers in deep models. Here, we display the fitting plot of one random layer in each model. Full results are given in Fig. C3–C13 in the Supplementary Material. (a) MLP on MNIST (layer fc2). (b) CNN on MNIST (layer Conv1). (c) CNN layers on CIFAR-10 (layer Conv3). (d) AlexNet on ImageNet (layer Conv3). (e) VGG-16 on ImageNet (layer Conv3). (f) ResNet-50 on ImageNet (layer res4b_branch2a). (g) DenseNet-121 on ImageNet (layer denseblock4_denselayer1_conv2). (h) EfficientNet-b1 on ImageNet (layer block_6_projectconv). (i) NF-Net-b1 on ImageNet (layer stage_11_attention_fc2). Naming of the layers follows that defined in Caffe or Pytorch.

To reduce the search space, we search $x_{\min}$ in the $\lfloor(n \times k\%)\rfloor$ smallest degree values, where n is the number of nodes in that layer, and $x_{\max}$ in the $\lfloor(n \times k\%)\rfloor$ largest degree values, respectively. Empirically, $k = 30$ is used. For each $(x_{\min}, x_{\max})$ pair, we estimate α using the method of maximum likelihood as in [67]. Following [79], we require that $\alpha > 1$.

To evaluate the goodness of fit, we compute the p-value by KS statistic. Since directly using KS statistic is unreliable if the parameters of the hypothesis depend on the data [82], [83], [84], [85], [86], we use the Monte Carlo simulations in [87] to compute the p-values. Specifically, a number of synthetic sample power law distributions with the same $(x_{\min}, x_{\max}, \alpha)$ as the fit model is generated using the inversion method [87]

$$x_i = \left\lfloor \frac{x_{\min}}{(1-(1-r^{\alpha-1})u_i)^{1/\alpha-1}} \right\rfloor$$

where $r = (x_{\min}/x_{\max})$, u_i is randomly sampled from the uniform distribution in [0, 1), and $\lfloor\cdot\rfloor$ denotes the floor operator. The p-value is defined as the fraction of KS distances obtained from the synthetic data that are larger than the empirical KS distance obtained from the observed data. We accept the hypothesis that the data are from the estimated TPL if $p > 0.05$ and reject it otherwise.

B. Empirical Evaluation on Deep Models

1) Existence of Power Law in DNNs: We have conducted power law fitting on the sparse degree distributions in the most popular DNN architectures and datasets (MLP on MNIST, LeNet-like CNN on MNIST & CIFAR-10, and AlexNet/VGG-16/ResNet-50/DenseNet-121/EfficientNet-b1/NF-Net-b1 on ImageNet [14], [15], [16], [88], [89], [90]) after pruning procedures [5]. Node degree accounts for the total number of connections (after pruning) to its upper and lower layers. For layers in MLP, the node degree of each neuron is the number of nonzero connections after the pruning. For a convolutional layer, we consider each feature map at the convolutional layer as a node. For instance, consider a feature map (node) in the first convolutional layer (conv1). As the input image is of size 28×28, each such feature map is of size 24×24. Unpruned networks illustrate the counting process here. We first count its connections to the input layer. Recall that in conv1, 1) the filter size is 5×5; 2) each filter weight is used $24 \times 24 = 576$ times; and 3) there is only one channel in the grayscale MNIST image. Thus, each node has $25 \times 576 \times 1 = 14\,400$ connections to the input layer. Similarly, for connections to the conv2 layer: 1) the conv2 filter size is 5×5; 2) each filter weight is used $8 \times 8 = 64$ times (size of each conv2 feature map); and 3) there are 32 feature maps in conv2. Thus, each conv1 node has $25 \times 64 \times 32 = 51\,200$ connections to the conv2 layer. Hence, the degree of each conv1 node (in an unpruned network) is $14\,400+51\,200 = 65\,600$. Note that the pooling layers do not have learnable connections. They are never sparsified, and we do not need to study their degree distributions. Details of the network architectures, corresponding accuracies, pruning, and power law fitting results are described in Section C-A in Supplementary Material C. Fig. 2 shows the CCDF plot for layers in the MLP, LeNet-like CNN, AlexNet, VGG-16, ResNet-50, DenseNet-121, EfficientNet-b1, and NF-Net-b1. The TPL fits the degree distribution well. Here, we only display one random layer from each deep model. Other results and details are listed in Section C-A in Supplementary Material C. Note that the work by Martin and Mahoney [91] also found the empirical spectral density (ESD) of weight matrices in well-trained DNNs displays power law distributions, which has several conceptual similarities with our findings. However, we have fully explored the internal connectivity of DNNs, while they analyzed the weight matrix.

2) Correlation Between DNNs Performances and Power Law Fitting Goodness: Sparse networks with different classification performances are explored for how well the TPL distribution fits them. The iterative pruning method is utilized as in [3]. We prune the dense networks by deleting 5% of the connections in each layer iteratively, starting from 95% and working up to 5% sparsity. Thus, we generate 19 sparse DNNs for each model.

We have performed MLP on MNIST, CNN on CIFAR-10, and AlexNet/VGG-16/ResNet-50/DenseNet-121/EfficientNet-b1/NF-Net-b1 on ImageNet. For each network, we select three layers from the top, middle, and bottom of the architectures, respectively. Thus, we have evaluated **152** sparse DNNs in total. The results (see Fig. 3) show that the TPL distribution fits the degree distribution better with higher p-values

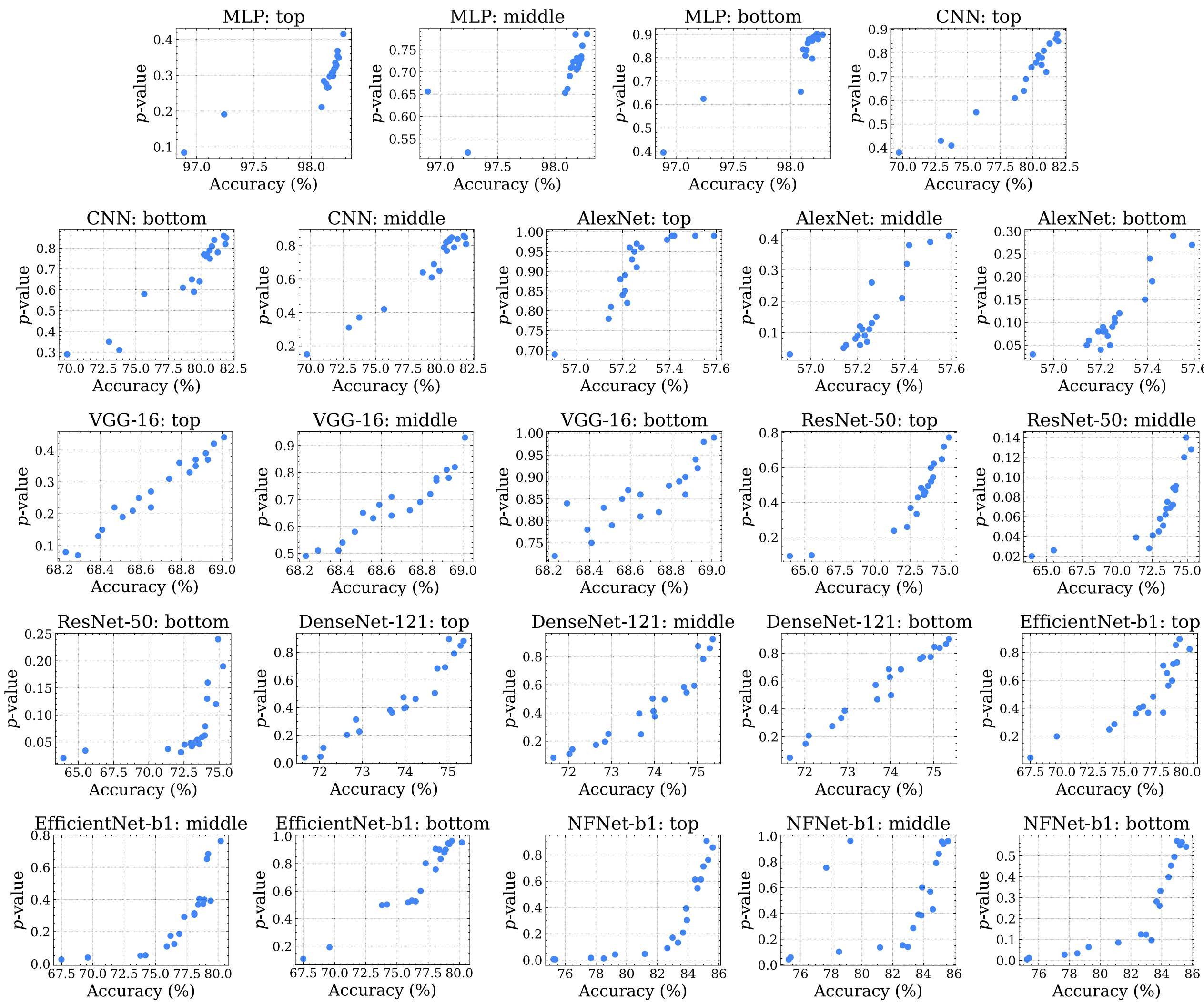

Fig. 3. p-values versus testing accuracies (%) for tested models in different tasks (MLP on MNIST, CNN on CIFAR-10, AlexNet on ImageNet, VGG-16 on ImageNet, and ResNet-50 on ImageNet). Each task has 19 models pruned by the method in [3]. For MLP, top: fc2, middle: fc1, and bottom: input; for CNN: top: fc3, middle: conv2, and bottom: conv1; AlexNet: top: fc8, middle: conv4, and bottom: conv1; VGG-16: top: conv5_3, middle: conv4_2, and bottom:conv5_2; ResNet-50: top: res5c_brach2c, middle: res3d_brach2c, and bottom: res2a_brach2c; DenseNet-121: top: denseblock4_denselayer14_conv1, middle: denseblock3_denselayer4_conv1, and bottom: denseblock1_denselayer3_conv1; EfficientNet-b1: top: block_22_projectconv, middle: block_5_projectconv, and bottom: block_1_conv; and NF-Net-b1: top: stage_36_conv3, middle: stage_21_attention_fc2, and bottom: stage_0_downsample_conv.

(goodness of fitting) in the layers in more well-trained networks. For those sparse networks with relatively low testing accuracies, the TPL distribution cannot fit in every layer, and the corresponding p-values are much lower. The results indicate a positive correlation between layers' goodness of fit for the power law distribution and the performance of the sparse networks.

Remark 1: TPL distribution fits better on layerwise degree distribution in sparse neural networks that perform well than in those that perform poorly (low testing accuracy). The TPL fitting level can be potentially used for verifying the generalization capability of sparse neural networks.

IV. Internal Preferential Attachment in Deep Neural Networks

In this section, we have proposed the preferential attachment model in DNNs' evolution. We further identify the existence of preferential attachment in continual learning tasks.

A. Preferential Attachment

Barabási and Albert [10] demonstrated that power law distributions can originate from preferential attachment, which can be either internal or external [9]. The model in [9] considers internal and external preferential attachment and assumes the following process: 1) nodes join the network at a constant rate; 2) new nodes connect to the existing nodes following external preferential attachment [see (6)]; and 3) already-existing nodes form new internal connections following internal preferential attachment [see (7)]. The aging of nodes is neglected. Moreover, all nodes and connections present in the network are active and able to initiate and receive new connections.

For external preferential attachment, $\Delta(d_1)$, the expected number of new connections for an existing degree-d_1 node satisfies

$$\Delta(d_1) = b\frac{d_1}{\sum_j d_j} \tag{6}$$

where b is the total number of new connections that new nodes create. Empirically, $\Delta(d_1)$ can be estimated as $\hat{\Delta}(d_1) = N(d_1)/D(d_1)$, where $N(d_1)$ is the number of new connections the new nodes create with existing degree-d_1 nodes, and $D(d_1)$ is the number of existing degree-d_1 nodes.

For internal preferential attachment, $\Delta(d_1, d_2)$, the expected number of new connections between two nodes with existing degrees d_1 and d_2 satisfies

$$\Delta(d_1, d_2) = 2Na \frac{d_1 d_2}{\sum_{s,m\neq s} d_s d_m} \tag{7}$$

where N is the total number of nodes in the network and a is the number of newly created connections per existing node per unit time. Empirically, $\Delta(d_1, d_2)$ can be estimated as

$$\hat{\Delta}(d_1, d_2) = N(d_1, d_2)/D(d_1, d_2) \tag{8}$$

where $N(d_1, d_2)$ is the number of new connections between an existing degree-d_1 node and an existing degree-d_2 node, and $D(d_1, d_2)$ is the number of existing node pairs: one with degree d_1 and the other with degree d_2.

B. Preferential Attachment in DNN Evolution

We propose internal preferential attachment in neural network evolution and study the properties of the sparsified networks that evolved via the preferential attachment model. In particular, we analyze some propositions of degree distribution evolution in internal preferential attachment. Assuming that only nodes in adjacent layers can be connected (a common assumption in feedforward neural networks), consider a pair of nodes: one from layer l with degree d_1 and the other from layer $(l+1)$ with degree d_2. If internal preferential attachment exists, the number of connections between this node pair grows proportional to the product $d_1 d_2$. Analogous to (7), the number of connections created per unit time between this node pair at time t is

$$\Delta_t^l(d_1, d_2) = N^l a^l \frac{d_1 d_2}{\sum_{s,m} d_s d_m} \tag{9}$$

where s and m are indices to all the nodes in the two layers involved, N^l is the number of nodes in layer l, and a^l is the number of new internal connections created per unit time from a node in layer l to a node in layer $(l+1)$. For node i at layer l, let $d_i(t)$ be its degree at time t. The following proposition shows how the degree evolves. In particular, $d_i(t)$ is linear with respect to the node's initial degree $d_i(0)$. Unlike in (7), note that there is no factor of 2 here.

Proposition 1:

$$\frac{\mathrm{d}d_i(t)}{\mathrm{d}t} = \frac{\left(N^l a^l + N^{l-1}a^{l-1}\right)d_i(t)}{\sum_s d_s(0) + (N^l a^l + N^{l-1}a^{l-1})t}$$

and $d_i(t) = d_i(0)c^l(t)$, where $c^l(t) = 1 + ((N^l a^l + N^{l-1}a^{l-1})t/\sum_s d_s^l(0))$. Proof can be found in Supplementary Material B.

Remark 2: Note that, as $c^l(t)$ increases with t, $d_i(t)$ increases with t. However, as we assume that new nodes cannot be added, the degree cannot grow infinitely, and $d_i(t)$ will not hold for large t.

Assume that the degree distribution of layer l at $t = 0$ (denoted p_0^l) follows the power law (standard or TPL); that is, $p_0^l(d) = Ad^{-\alpha}$ for some $A > 0$ and $\alpha > 1$. Then, the following proposition shows the closed-form solution of the degree distribution $p_t^l(d)$ of layer l at time t.

Proposition 2: $p_t^l(d) = (c^l(t))^{\alpha} Ad^{-\alpha}$, which follows the same power law as p_0^l, but scaled by the factor $(c^l(t))^{\alpha}$. Proof can be found in Supplementary Material B.

Corollary 1: For any layer, if its initial degree distribution follows the power law, its degree distribution at time t also follows the same power law scaled by a certain factor.

C. Existence of Power Law and Preferential Attachment in Continual Learning

To study the dynamic properties of neural network connectivity, we consider the continual learning setting in which consecutive tasks are learned [50]. Here, we identify the existence of power law dynamics in this process.

The continual learning problem with two tasks on the MNIST dataset from [50] and [92] is investigated. Task A used the original images, while Task B used images in which pixels inside a central $P \times P$ square are permuted, where $P = 8$.

We first train a dense network on Task A as in Fig. 4. A fraction of $s_1 = 0.9$ connections is pruned from each layer. Connections to the last softmax layer are not pruned. The remaining connections are retrained on Task A and then frozen. Next, we reinitialize the pruned connections and trained a dense network on Task B. Another fraction of $s_2 = 0.8$ connections from each layer is again pruned, and the remaining connections are retrained.

Because the connections learned on Task A are frozen, they are not pruned or retrained. The structure of the MLP model here is input $-$ 1,024FC $-$ 1,024FC $-$ 10softmax. Testing accuracies of dense networks for Tasks A and B are 98.09% and 98.17%, and those of sparse networks for A and B are 98.21% and 98.09%, respectively. The sparse network exhibited good performance on both tasks.

1) Existence of the Truncated Power Law: We have verified that the degree distribution of the sparse MLP model trained on Task A follows the TPL. We have found that its degree distribution after training on Task B also follows the TPL with the p-values of 0.200, 0.960, and 0.436 for input, fc1, and fc2 layers, respectively (all larger than 0.05). The p-value of the first layer is relatively low because, due to permutation, the input layer has to establish new associations to map from pixels to penstrokes.

2) Existence of Internal Preferential Attachment: Here, we verify whether there existed internal preferential attachment in the continual learning process. Empirically, the number of new connections between two nodes with existing degrees d_1 and d_2 $\Delta_t^l(d_1, d_2)$ [details can be found in (9)] is estimated by counting the number of new connections created at time $(t+1)$ between all the involved node pairs (i.e., one from layer l with degree d_1 and one from layer $(l+1)$ with degree d_2 at time t) and then dividing it by the number of such node pairs.

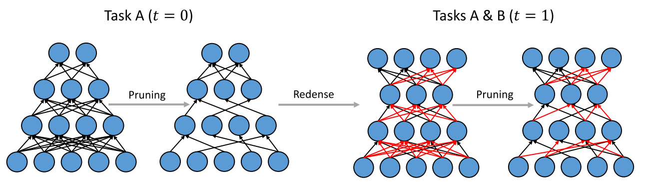

Fig. 4. Continual learning setup. Connections learned for Task A are in black, and those learned for Task B are in red.

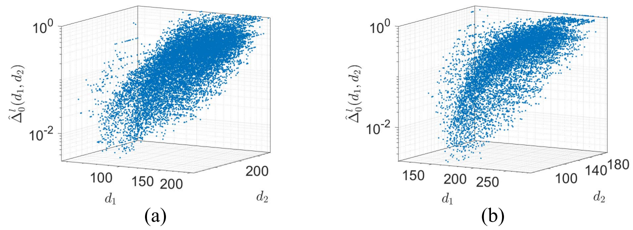

Fig. 5. $\hat{\Delta}_0^l(d_1, d_2)$ versus d_1 and d_2. (a) Input-to-fc1. (b) fc1-to-fc2.

Fig. 5 shows this empirical estimate at time $t = 0$ [denoted $\hat{\Delta}_0^l(d_1, d_2)$] versus the degrees d_1 and d_2. As can be seen, $\hat{\Delta}_0^l(d_1 d_2)$ increased with d_1 and d_2, indicating the presence of internal preferential attachment.

V. Power Law Dynamics in Sparse Network Generation and Continual Learning

With the discovery of TPL distribution and preferential attachment in DNNs, we propose to generate compact and well-performing DNNs from a topology-driven perspective. Specifically, two applications are demonstrated, including sparse network generation and network growth in continual learning using preferential attachment in DNNs.

A. Sparse Networks Generated by Preferential Attachment

We have developed a novel sparse network generation method, which utilizes preferential attachment to redense the pruned networks. One major issue facing the magnitude-based pruning in [5] is the possibility of irretrievable network damage [21]. In the pruning procedure, incorrect pruning or the elimination of wrong weights may result in accuracy loss because certain connections with small weights may be also critical to network performance [21], [93]. To tackle this challenge, we use preferential attachment as a postpruning procedure to generate new sparse DNNs by relinking some pruned connections. To be more specific, the proposed network generation method includes two operations: three-step pruning and preferential attachment to redense the network. The redensing here functioned as a splicing operation, which can relink the incorrectly pruned connections to achieve network recovery. Different from [21], their proposed method conducts splicing based on the evaluation of connection importance, which is time- and computation-consuming. Our approach is simple and efficient since it directly utilized the TPL distribution in the network topology.

As shown in Fig. 6, we first prune a dense (trained) network and obtain an initial sparse network with connection fraction CF $= s_1\%$ using the pruning method in [5]. Second,

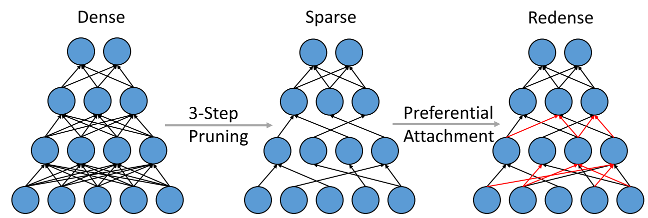

Fig. 6. Sparse network generation setup.

we utilize preferential attachment to generate connections (redense process) to achieve a network with CF $= s_2\%$, where $0 < s_1 < s_2 < 100$. Finally, we retrain this using the same initialization method in [3]. We only consider internal preferential attachment as in (9). We test that the TPL fits well in all layers of these models with CF $= s_1\%$. Thus, the TPL can also fit the degree distribution in layers of the redensed networks with CF $= s_2\%$ using preferential attachment due to Corollary 1. The baselines are given in the following:

1) initial pruned networks in [5];
2) uniform attachment strategy, with all equal probability $(1/N^l)$ for each node to add connections in layer l with N^l nodes and CF $= s_2\%$;
3) three-step pruning methods in [5] with CF $= s_2\%$, which is performed on the basis of the original dense model;
4) dynamic network surgery method in [21];
5) using AdamW [94] to populate the weights with CF $= s_2\%$;
6) using Taylor [31], one of the representative structured pruning methods, pruning the model to CF $= s_2\%$;
7) dense models (CF $= 1.0$).

We perform the experiments of MLP (input-1024FC-1024FC-10softmax) on MNIST, CNN (input $\rightarrow 32C3 \rightarrow 32C3 \rightarrow$ MP2 $\rightarrow 64C3 \rightarrow 64C3 \rightarrow$ MP2 $\rightarrow$ 512FC $\rightarrow$ 10softmax) on CIFAR-10, and AlexNet/VGG-16/ResNet-50 on ImageNet for evaluation [95]. For initial models of AlexNet and VGG-16, we use the pruned models provided in [5]. For initial models of MLP, CNN, and ResNet-50, we prune them following the three-step pruning method since they are not available in [5]. For dense models, including MLP, AlexNet, VGG-16, and ResNet-50, we use the models provided by Caffe Model Zoo. For dense CNNs, we train by ourselves.

The results on accuracy and CFs have been illustrated in Table II. Previously, dynamic network surgery in [21] only provided experiments of AlexNet on ImageNet. In their results, CF $= 0.06$, and top-one accuracy is 56.91%. Table III compares the FLOPs (normalized to three-step pruning) among all pruning-based methods, including three-step pruning, preferential attachment, and Taylor. Our work finds the following.

1) Preferential attachment functions, as a postpruning splicing operation, can boost network accuracy compared with the initial networks with the pruning procedure only in [5].
2) Sparse networks generated by preferential attachment outperform those using uniform attachment among all models.

TABLE II

VALIDATION ACCURACIES FOR DIFFERENT MODELS IN SPARSE NETWORK GENERATION EXPERIMENTS. THE BEST AND SECOND BEST PERFORMANCES ARE MARKED IN BOLD RED AND BLUE, RESPECTIVELY

Networks	MLP		CNN		AlexNet		VGG-16		ResNet-50	
	CF	Top 1 (%)	CF	Top 1 (%)	CF	Top 1 (%)	CF	Top 1 (%)	CF	Top 1 (%)
Initial in [5]	0.10	98.21	0.30	80.46	0.11	57.23	0.07	68.66	0.15	72.29
Preferential attachment		**98.23**		**81.79**		**57.26**		**69.15**		73.47
Uniform attachment		**98.23**		80.87		57.03		67.31		72.33
3-step pruning [5]	**0.30**	**98.24**	**0.50**	81.32	**0.25**	57.22	**0.30**	68.93	**0.30**	**73.50**
AdamW [94]		98.22		81.50		57.13		68.72		72.98
Taylor [31]		98.19		81.64		**57.24**		**68.96**		73.29
Dense	1.00	98.09	1.00	**81.89**	1.00	57.22	1.00	68.50	1.00	**75.99**

TABLE III

NORMALIZED FLOPS FOR DIFFERENT MODELS IN SPARSE NETWORK GENERATION EXPERIMENTS

Networks	MLP	CNN	AlexNet	VGG-16	ResNet-50
3-step pruning [5]	1.00	1.00	1.00	1.00	1.00
Preferential attachment	1.51	1.47	1.39	1.26	1.25
Taylor [31]	2.02	3.14	3.26	3.58	4.07

3) We have more FLOPs than three-step pruning since we add connections on the basis of the three-step pruning. However, as shown in Table III, our method can have fewer FLOPs than Taylor, which is one of the representative approaches in network pruning. This indicates that, although we further evolve connections after pruning, our method is more computational-efficient than methods only pruning the networks.
4) In MLP/AlexNet/VGG-16, preferential-attachment-evolved networks have higher accuracies than dense networks with lightweight network structures. Other models (CNN and ResNet-50) share similar accuracies of preferential attachment networks and dense networks, exhibiting only a 0.10% gap for CNN and a 2.49% gap for ResNet-50;
5) In CNN/AlexNet/VGG-16, networks evolved by preferential attachment are more accurate than those by the three-step pruning method in [5] with the same CF values. For MLP and ResNet-50, the accuracies of preferential-attachment-evolved networks and three-step-pruned networks are comparable, in which the gaps are only 0.01% and 0.03% for MLP and ResNet-50, respectively. This is because MLP and ResNet-50 already learn powerful representations on MNIST and ImageNet, respectively, where the room for improvements could be limited.
6) Using preferential attachment can achieve higher accuracy than other pruning methods, including the dynamic network surgery method in [21] and Taylor [96], and directly using AdamW [94] to populate weights.

More importantly, networks evolved by preferential attachment have fewer parameters (as shown by the CF: a larger CF indicates that more parameters are needed). For deep CNNs, such as AlexNet/VGG-16/ResNet-50, the preferential attachment method can save about 70% storage without performance degradation compared with corresponding dense models.

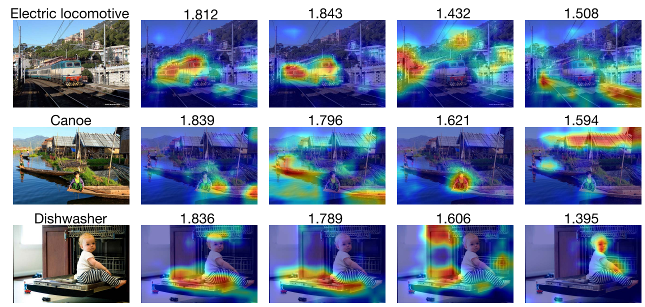

Fig. 7. Visualization on the channels from the last residual layer in sparse ResNet-50 generated via preferential attachment. The numbers of connections (10^4) are given on each visualized image.

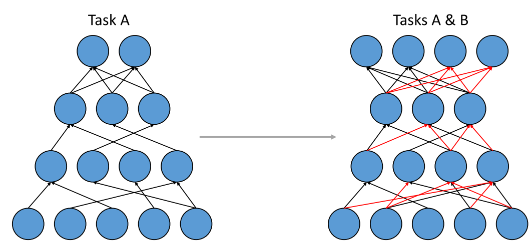

Fig. 8. Illustration of modified PathNet with lateral connections chosen by preferential attachment.

The improvement is generally caused by the mechanism of preferential attachment, where the nodes with higher degrees are more encouraged to evolve new connections than others. In the initial sparse network, those nodes with high degrees are mostly critical channels/neurons in DNNs. Fig. 7 visualizes the channels from the last residual layers in generated ResNet-50 trained on ImageNet via using Grad-CAM in [97]. The average channel degree is 1.682×10^4. We can find that those critical channels (related to the key semantics) are with higher degrees. Those channels with fewer degrees are mostly corresponding to some less critical representations. For example, in the first case in Fig. 7, the ground-truth label is an electric locomotive. Those channels with smaller numbers of degrees (columns 4 and 5) are activated to the trees, houses, and rail tracks, which are not related to the ground-truth label, while those with high degrees (columns 2 and 3) are activated to key features of the electric locomotive. Preferential attachment can progressively enhance these critical channels.

In short, our method is simple, efficient, and capable to retain what can be learned. Using the derived lightweight networks can cut down the inference time and storage burden.

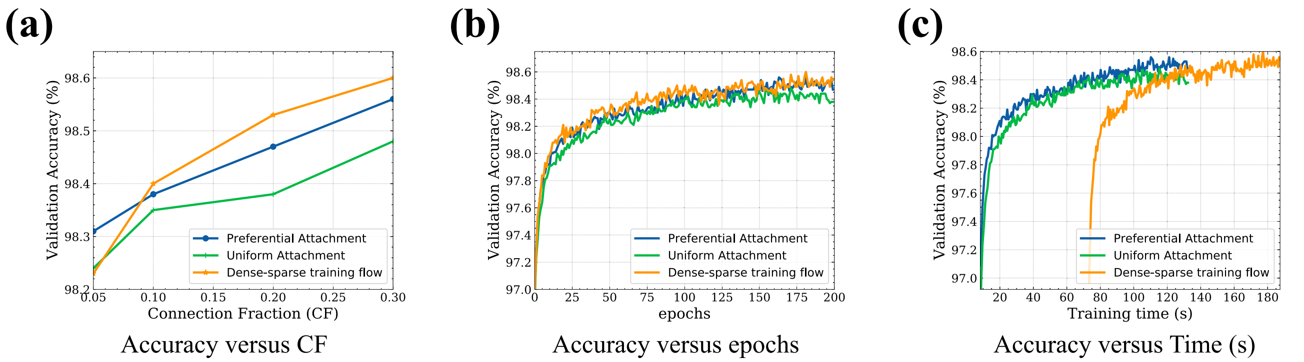

Fig. 9. Validation accuracy (%) for MLP on MNIST with permutation size $P = 8$ and with K=400. (a) Validation accuracy versus different CFs. (b) Validation accuracy versus training epochs. (c) Validation accuracy versus training time. In (b) and (c), CF = 0.3.

B. Using Preferential Attachment in Continual Learning

In biological neural networks, the reuse of neural circuits is essential for learning various tasks within a limited capacity. Scale-free networks are shown to resemble biological neural networks most and achieve the fastest learning among many network topologies. Analogously, within a single artificial neural network, we also expect to speed up the continual learning process and obtain more economical representations through the exaptation of the learned connectivity. With the identification of preferential attachment in the continual learning process in Section IV-C, we propose to evolve sparse DNNs with preferential attachment to adapt to a sequence of tasks.

We experiment on two kinds of settings, considering both internal and external preferential attachment. The first setting is like the structure of PathNet [57], where the number of nodes is fixed, and only new connections are created; the second setting is like the structure of progressive neural networks (PNNs) [55], where not only new nodes are added but also new connections are added.

1) Speeding Up Training With Preferential Attachment: We use a structure similar to that of PathNet [57]. Instead of using a modular network where each module can be either a convolutional or a linear neural network as in PathNet, we simply use the node as the unit in the experiment.

We use the MNIST dataset and the same two tasks that are defined in [54] with the same training/validation/testing split. We decrease the sparsity of the network when a new task arrives and see whether adding connections to the previous hidden-layer activations following the preferential attachment process helps the network to learn the second task faster than a uniform attachment. The model can be seen in Fig. 8. The first column shows a learned sparse network, where the black connections are those that remained after pruning. The second column (see the added two nodes in the top layer in the right part of Fig. 8) shows the new sparse network learned on Task B, where the red connections are newly added. Specifically, we first train a sparse neural network with $\mathsf{K} = 400$ nodes in each of the two hidden layers, trained on Task A by pruning the unimportant connections with small magnitudes in each layer: input-KFC-KFC-10softmax. The sparsity for each layer is $s = 0.9$. Then, we fix the learned connections and switch to Task B. A fraction of CF $\in \{0.1, 0.2, 0.3\}$ new connections in each hidden layer is generated via the following:

1) internal preferential attachment,
2) uniform attachment;
3) dense-sparse training flow in Fig. 4.

Fig. 9(a) shows the accuracy results over different CFs for MLP with $\mathsf{K} = 400$ when $P = 8$. Recall that P is the permutation size. The fraction of added connections is chosen from CF = $\{0.1, 0.2, 0.3\}$. Given the same number of added connections, generating new connections with preferential attachment achieves higher validation accuracy than uniform attachment in all cases. For both preferential attachment and uniform attachment, accuracy continued to grow as the CF increases. Preferential attachment achieves results similar to those of the dense-sparse training flow across all CFs [see Fig. 9(a) and (b)] while being much cheaper in training [see Fig. 9(c)]. We use masking operations to deal with sparse weights. More speedup can be obtained using targeted operations for sparse weights. The dense-sparse training flow results in better accuracy than preferential attachment except when CF = 0.05. However, directly generating new connections based on preferential attachment is much more efficient than using the dense-sparse training flow in Fig. 4 as this method only needs fine-tuning a sparse network instead of first training a dense network and then fine-tuning the pruned network.

2) Better Learning With Fewer Training Samples: Inspired by humans' ability to learn a second task faster with fewer training examples after a first task learned to a certain accuracy, we have constructed a similar scenario using the MNIST dataset in [54] where the second task contains only 500 out of the original 50 000 training samples. Fig. 10a shows accuracy versus different CFs in this case. When the network size is large ($\mathsf{K} = 2048$), and the training sample size is small, the performances of the three methods drop as CF increases [see Fig. 10(a)]. This is because, when the capacity is large, the methods tend to overfit. Unlike previous findings, choosing connections using preferential attachment has the best performance under all CFs. The dense-sparse training flow performs the worst because the network easily overfits during the redense process with limited data. On the other hand, preferential/uniform attachment, which does not have the redense process and directly select limited connections, and is less overfitted. Meanwhile, preferential attachment performs

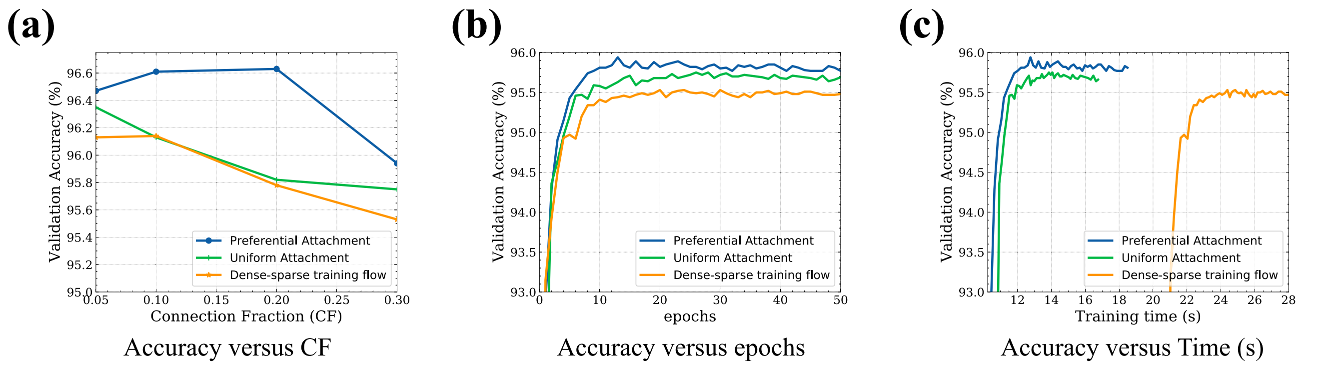

Fig. 10. Validation accuracy (%) for MLP on MNIST with permutation size $P = 8$ and with $\mathsf{K} = 2{,}048$ under few-sample experimental settings. (a) Validation accuracy versus different CFs. (b) Validation accuracy versus training epochs. (c) Validation accuracy versus training time. In (b) and (c), $\mathrm{CF} = 0.3$.

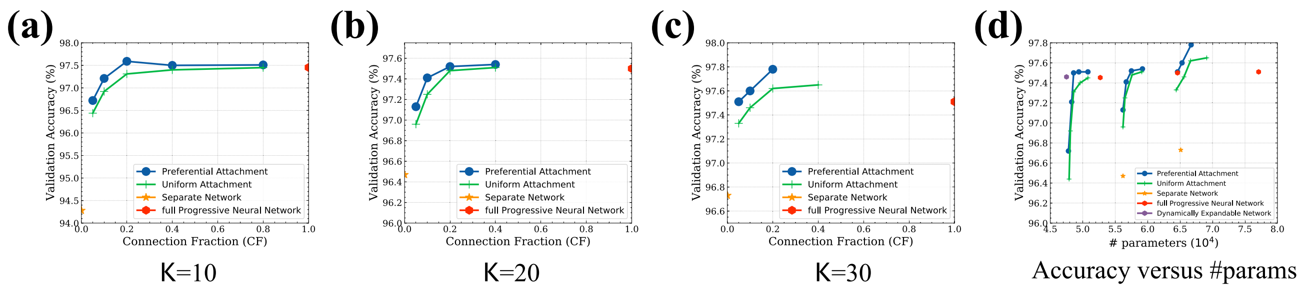

Fig. 11. Experimental results of MLP on MNIST with the PNN structures. (a)–(c) Validation accuracy (%) versus added capacity for MLP on MNIST with permutation size $P = 8$. (d) Validation accuracy (%) versus #parameters for MLP on MNIST with permutation size $P = 8$.

consistently better and faster [see Fig. 10(b) and (c)] than uniform preferential attachment because it leverages prior connectivity knowledge from the first task.

3) Storage Saving With Preferential Attachment: Using the structure of PNNs [55], we first use the method in [5] and train a single column for the Task A: a deep sparse neural network having L layers with hidden activations $h_l^{(1)} \in \mathbb{R}^{N_l}$ at layer $l \leq L$, by pruning the unimportant weights with small magnitudes in each layer. Then, we switch to Task B, the parameters $\Theta^{(1)}$ for Task A are fixed, and a new column with parameters $\Theta^{(2)}$ is instantiated (with random initialization), where layer $h_l^{(2)}$ receives input from both $h_{l-1}^{(2)}$ and $h_{l-1}^{(1)}$ via a fraction of CF lateral connections, generated using an external preferential attachment or uniform attachment. Suppose that there are N_{l-1} nodes in the $(l-1)$th layer of the first column, and the degree of the jth node is $d_{l-1}^{(1)}(j)$. We vary the CF from $\{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1\}$. We compare the performances of these methods.

1) *Separate Networks (CF = 0):* Train a separate network on Task B, without any lateral connection to the first column.
2) *Uniform Attachment ($0 < CF < 1$):* Generate lateral connections with preferential attachment. Specifically, one node in the lth layer of the second column is connected to node j of the $(l-1)$th layer of the first column with probability $((d_{l-1}^{(1)}(j))/(\sum_{k=1}^{N_{l-1}} d_{l-1}^{(1)}(k)))$. For node j, as we only consider its lateral connections to the upper layer in the second column, its degree is calculated by only counting connections to its upper layer in the first column.
3) *Preferential Attachment ($0 < CF < 1$):* One node in the lth layer of the second column is connected to each node of the $(l-1)$th layer with the same probability $(1/(N_{l-1}))$.
4) *Full PNNs (CF = 1):* All the newly added nodes $h_l^{(2)}$ in layer i are fully connected to $h_{l-1}^{(1)}$.
5) *DENs [56]:* Three-step learning for new tasks: performs selective retraining, dynamically expands network capacity, and prevents semantic drift by splitting/duplicating units.
6) *Structured Continuous Sparsification (SCS) [98]:* It continually grows the networks via structured masks for each new task. SCS is initially proposed in the field of neural architecture search (NAS). However, since the core idea of continuous growth behind it is conceptually similar to ours, we compare it here. We only compare it using deep models (VGG-16 and ResNet-18) since their original work is working toward the deep models.

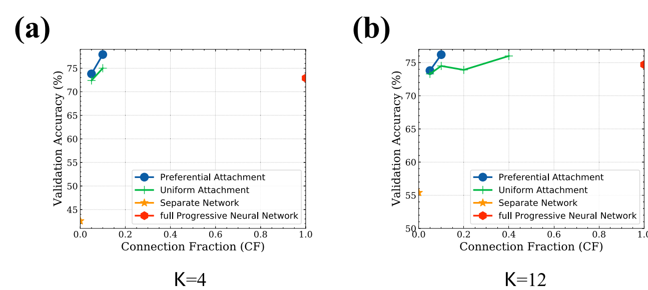

Fig. 12. Validation accuracy (%) versus added capacity for CNN on CIFAR-10 with the PNN structure. (a) and (b) Validation accuracies versus different CFs with $\mathsf{K} = 4$ and $\mathsf{K} = 12$.

We use the MNIST, CIFAR-10/100, and MiniImageNet datasets. For each dataset, we use the same two tasks as

TABLE IV

ACCURACY ON SPLIT CIFAR-100 TASKS. THE BEST PERFORMANCES EXCLUDING THE SEPARATE NETWORKS ARE MARKED AS **BOLD**. HERE, UA AND PA DENOTE UNIFORM AND PREFERENTIAL ATTACHMENT, RESPECTIVELY

Methods	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Separate	66.2	79.2	74.9	79.8	83.5	82.9	81.0	82.3	85.1	86.2	85.4	81.7	83.6	78.8	52.9	71.4	65.4	71.3	85.9	89.3
PNN	66.9	74.5	74.9	76.3	82.4	82.1	79.1	78.6	79.5	83.9	84.6	79.2	83.1	81.0	49.9	69.6	65.8	69.7	85.3	86.9
DEN	66.1	77.4	76.2	77.9	83.1	82.4	79.6	79.5	81.1	84.7	86.4	79.0	82.9	81.6	51.5	70.7	67.2	**79.5**	86.1	89.8
UA	65.2	72.7	74.8	73.5	76.8	77.9	76.2	78.5	78.6	81.9	80.3	75.1	81.0	77.7	46.5	69.5	63.4	65.1	83.2	87.6
SCS	66.9	**77.8**	75.6	80.3	81.0	83.2	78.5	79.9	81.3	84.2	84.5	81.2	82.4	79.2	51.0	70.3	64.2	67.9	84.8	88.5
PA	**67.3**	77.2	**78.1**	**82.4**	**84.8**	**84.9**	**80.3**	**81.7**	**82.6**	**86.6**	**88.3**	**82.5**	**84.1**	**82.7**	**52.6**	**72.4**	**69.3**	78.0	**87.6**	**91.4**

defined in [47], [54], and [99] with the same training/validation/testing split. Instead of decreasing the sparsity of the network when a new task arrived, we add new nodes (or feature maps for CNN) to the hidden layers for the second task. The models for the experiments are shown in Fig. D14 in the Supplementary Material.

a) MLP on MNIST: The first column is a sparse neural network with two hidden layers, with $\mathsf{K} = 200$ nodes in each hidden layer. The sparsity for each layer is $s = 0.8$. Then, we build a second narrow column for Task B, with only $\mathsf{K} \in \{10, 20, 30\}$ hidden units in each hidden layer. Fig. 11(a)–(c) shows the accuracy results over different CFs when K varies. Only CFs smaller than or equal to the one that gives the highest validation accuracy are shown. As can be seen, using preferential attachment consistently achieves better accuracy than using a uniform attachment. With only 5% of the lateral connections to Task A, preferential attachment achieves significantly better accuracy than training a separate network. Moreover, with around 20% added capacity, preferential attachment achieves results comparable to or even better than those of the full PNNs. Though DEN has the smallest number of parameters [see Fig. 11(d)], its accuracy is significantly worse than using a preferential attachment with only a slightly larger number of parameters. On the other hand, DEN performs an expensive three-step connection selection process, while preferential attachment cheaply and directly selects connections according to probability.

b) CNN on CIFAR-10: The first column is a CNN architecture for CIFAR-10 classification: *input-32C3-32C3-MP2-64C3-64C3-MP2-512FC-10softmax.* The sparsity for each convolutional layer is $s = 0.7$. Then, we build a second narrow column for Task B with $\mathsf{K} \in \{4, 12\}$ feature maps in the first two convolutional layers and $2\mathsf{K}$ feature maps in each of the following two convolutional layers, and the number of nodes in the FC layers is unchanged. Fig. 12(a) and (b) shows the accuracy results over different CFs when K varied. Only CFs smaller than or equal to the one that gives the highest validation accuracy are shown. Results suggest that preferential attachment consistently achieves better accuracy than uniform attachment. With only 5% of the lateral connections to Task A, preferential attachment achieves significantly better accuracy than training a separate network. Moreover, with around 10% added capacity, preferential attachment achieves results comparable to or even better than the full PNNs.

c) VGG-16 on Split-CIFAR-100: We have divided the CIFAR-100 dataset into 20 tasks, where each task includes five classes, and there are no overlapping classes among different tasks. We follow the experimental setups in [56]. Specifically, each task has 2500 training samples and 500 testing samples. We utilize a VGG-16 as the baseline model for the first task. The network will expand with CF $= 0.1$ when a new task comes in. We evaluate the testing accuracy of each task after training all tasks sequentially. We compare with the structure-based approaches, including PNN and DEN. In addition, we also evaluate the separate networks that train each task from scratch separately, for comparison. No forgetting can occur in the separate model, which can serve as the upper bound for the continual learning task. Table IV shows that our method can consistently perform better against PNN and DEN in most tasks. Even within tasks, the performance of our method can outperform the separate model, which verifies the existence of forward transfer across tasks in our model. Tables V and VI give the relative model size and FLOPs on 20 tasks, showing that our model has the smallest model size and FLOPs among the baselines.

TABLE V

COMPARISON ON FINAL MODEL SIZE (NORMALIZED TO THE SIZE OF OUR MODEL)

Methods	Ours	PNN	DEN	SCS
CIFAR-100	1.0	6.9	1.2	1.5
Mini-ImageNet	1.0	7.0	1.7	1.3

TABLE VI

COMPARISON ON FLOPS (NORMALIZED TO THE SIZE OF OUR MODEL)

Methods	Ours	PNN	DEN	SCS
CIFAR-100	1.0	2.6	4.4	5.9
Mini-ImageNet	1.0	2.3	4.6	6.2

d) ResNet-18 on Split-MiniImageNet: To verify the effectiveness of using preferential attachment in network evolution with high-resolution datasets, MiniImageNet [47], [99], [100] was split to 20 tasks, each with five classes. Similarly, we train the full network in the first task and then grow the network for each task with CF $= 0.15$. We compare with PNN, DEN, and the separate models. Table VII again shows that our method can outperform PNN and DEN in most tasks (18 out of 20). Tables V and VI show that our model can significantly save model size and computational burden compared with the baselines.

In a nutshell, under continual learning settings, where the network needs more capacity to accommodate more knowledge for new tasks, preferential attachment helps to choose useful lateral connections that can lead to smaller networks and faster learning but require fewer samples. Thus, the

TABLE VII

ACCURACY ON MINIIMAGENET SPLIT 20 TASKS. THE BEST PERFORMANCES EXCLUDING THE SEPARATE NETWORKS ARE MARKED AS **BOLD**. HERE, UA AND PA DENOTE UNIFORM AND PREFERENTIAL ATTACHMENT, RESPECTIVELY

Methods	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Separate	65.3	68.5	64.7	61.9	69.2	74.1	65.8	63.0	61.9	67.3	63.8	67.6	62.9	69.6	67.8	68.4	71.3	65.6	67.8	64.9
PNN	62.6	60.9	61.8	59.6	62.0	66.3	59.8	57.9	59.2	58.3	59.1	62.5	54.3	63.2	61.2	58.5	61.3	57.6	63.5	58.9
DEN	64.6	65.3	63.8	61.1	64.6	67.9	61.6	60.9	59.0	**60.2**	59.5	63.8	54.9	**65.3**	62.9	59.9	**65.2**	58.7	65.1	59.4
UA	57.9	58.4	61.0	56.3	61.1	59.6	53.9	57.2	54.3	57.8	55.6	60.3	52.1	56.5	58.4	57.0	59.8	53.2	59.4	53.1
SCS	65.6	64.3	62.5	57.9	62.0	64.8	60.7	56.5	56.9	58.0	56.3	62.5	54.8	61.5	62.2	58.8	62.0	57.6	62.6	58.7
PA	**65.9**	**66.7**	**64.9**	**62.0**	**65.3**	**69.3**	**62.9**	**62.1**	**61.7**	**60.2**	**61.3**	**65.3**	**57.4**	65.0	**64.8**	**63.3**	64.7	**62.9**	**66.2**	**61.2**

proposed framework is well-suited for tackling the challenges of both latency and accuracy in continual learning. On the one hand, we can reduce the inference time by using lightweight networks with few-sample learning ability. On the other hand, we can ensure the good performance of AI systems by the effectiveness of power law topology in DNNs.

VI. CONCLUSION

In this article, we have identified that the connectivities inside sparse DNNs follow the variants of power law distribution, which can be observed in biological or social networks. Specifically, the degree distribution of pruned networks obeys the TPL, while the network performances and the goodness of power law fitting are positively correlated. Preferential attachment, which can generate networks with power law distribution, is also found in the process of continual learning. Based on the empirical findings, we have proposed novel approaches to sparse network generation and continual learning, enabling real-world applications that require high-performance and lightweight DNNs. In future work, we plan to explore our findings with other deep learning backbones (e.g., vision transformers) and design specific low-level matrix operation and hardware implementation for our sparse network generation approach.

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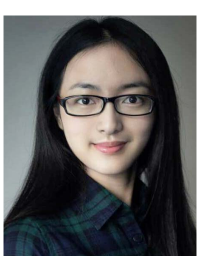
Fan Feng is currently pursuing the Ph.D. degree with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong.

His research interests include the interdisciplinary areas of machine learning and complex networks, aiming at advancing continual/lifelong machine learning algorithms using graphical models.



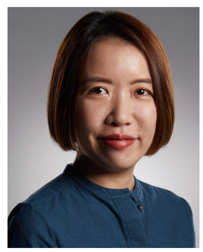
Lu Hou received the Ph.D. degree from the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2019.

She is currently a Senior Researcher at the Huawei Noah's Ark Lab, China. Her research interests lie in developing efficient deep learning models with lower memory and computation costs, especially for large pretrained language models.

Dr. Hou received the Outstanding Paper Award at the Annual Meeting of the Association for Computational Linguistics (ACL) 2022.

Qi She worked at Intel Labs, China, focusing on continual learning, computer vision, and robotics. He is currently a Research Scientist on machine learning on ads model with Bytedance AI Lab, Beijing, China. He has more than 40 peer-reviewed publications, including the International Conference on Machine Learning (ICML), the Uncertainty in Artificial Intelligence (UAI), the Conference on Computer Vision and Pattern Recognition (CVPR), the International Conference on Computer Vision (ICCV), the International Conference on Learning Representation (ICLR), the AAAI Conference on Artificial Intelligence (AAAI), the IEEE International Conference on Robotics and Automation (ICRA), the British Machine Vision Conference (BMVC), the *Artificial Intelligence*, the IEEE TRANSACTIONS ON SIGNAL PROCESSING (TSP), and the *ACM Computing Survey* (CSUR). He holds five granted/filed U.S. patents and 20 China patents.

Dr. She is a PC Member of the International Conference on Neural Information Processing (ICONIP) 2019. He serves as a Reviewer for prestigious conferences and journals, including the Conference on Neural Information Processing System (NeurIPS), ICML, ICLR, CVPR, AAAI, the International Joint Conference on Artificial Intelligence (IJCAI), ICONIP, TSP, CSUR, and *European Journal of Neuroscience* (EJN). He was the Organizer of the IEEE International Conference on Intelligent Robots and System (IROS) 2019 Lifelong Robotic Vision Challenge and the CVPR 2020 & 2021 Continual Learning in Computer Vision Workshop.



Rosa H. M. Chan (Senior Member, IEEE) received the B.Eng. degree (Hons.) in automation and computer-aided engineering from The Chinese University of Hong Kong, Hong Kong, in 2003, and the M.S. degree in aerospace engineering and the Ph.D. degree in biomedical engineering from the University of Southern California, Los Angeles, CA, USA, in 2011.

She is currently an Associate Professor and an Associate Head (Research and Graduate Programmes) of the Department of Electrical Engineering, City University of Hong Kong, Hong Kong. Her research interests include computational neuroscience, neural prosthesis, and human–computer interaction.

Dr. Chan is an Editorial Board Member of the *Journal of Neural Engineering* and an Area Editor of *Encyclopedia of the Human Brain* (IEEE-Wiley). She was a co-recipient of the Outstanding Paper Award of IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING in 2013 for the research breakthroughs in mathematical modeling for the cognitive prosthesis. She was the Chair of the Hong Kong-Macau Joint Chapter of the IEEE Engineering in Medicine and Biology Society (EMBS) in 2014. She is elected to the IEEE EMBS Administrative Committee (AdCom) as an Asia Pacific Representative for the term 2018–2023.

James T. Kwok (Fellow, IEEE) received the Ph.D. degree in computer science from The Hong Kong University of Science and Technology, Hong Kong, in 1996.

He is currently a Professor with the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology. His research interests include machine learning, deep learning, and artificial intelligence.

Dr. Kwok is serving as an Editorial Board Member for the *Machine Learning*. He received the IEEE Outstanding 2004 Paper Award and the Second Class Award in Natural Sciences by the Ministry of Education, China, in 2008. He has served/is serving as the senior area chair/area chair of major machine learning/AI conferences, including the Conference on Neural Information Processing System (NeurIPS), the International Conference on Machine Learning (ICML), the International Conference on Learning Representation (ICLR), the International Joint Conference on Artificial Intelligence (IJCAI), the AAAI Conference on Artificial Intelligence (AAAI), and the European Conference on Machine Learning (ECML). He is serving as an Associate Editor for the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the *Neural Networks*, the *Neurocomputing*, the *Artificial Intelligence Journal*, and the *International Journal of Data Science and Analytics*.